Session 07 (AIML) - Assignment Questions

Q1. (Creating Tuples)
Write a program to create tuples using different methods:
a) A tuple with 5 numbers using parentheses.
b) A mixed tuple containing integer, string, float, and boolean.
c) An empty tuple using two different ways.
d) A single-element tuple with the value 99.

Print all tuples and their types. Add comments explaining the rules for single-element tuples.

Q2. (Tuple Packing)
Create a tuple without using parentheses (tuple packing) to store: your name, age, and city.
Print the tuple and its type.
Then unpack the tuple into three separate variables and print them individually.

Q3. (Indexing and Negative Indexing)
Create the tuple: colors = ('red', 'green', 'blue', 'yellow', 'purple', 'orange')
Print:

- First element
- Third element
- Last element (using negative index)
- Second last element (using negative index)

Take an index number as input from the user and print the element at that index.

Q4. (Slicing Tuples)
Given the tuple: numbers = (0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10)
Using slicing, print:
a) Elements from index 2 to 7
b) First 5 elements

c) Last 4 elements
d) Every second element
e) The entire tuple in reverse order

Add comments explaining the slicing syntax.

Q5. (Nested Tuples)
Create a nested tuple to represent a 3×3 matrix:
matrix = ((1, 2, 3), (4, 5, 6), (7, 8, 9))

Print:

- The first row
- The element at second row, third column
- The element at third row, first column

Q6. (Iterating and Unpacking)
Create a tuple: student = ('Rahul', 20, 'Computer Science', 'A')
a) Iterate through the tuple using a for loop and print each item.
b) Unpack the tuple into four variables (name, age, branch, grade) and print them with descriptive messages.

Q7. (count() Method)
Create the tuple: grades = ('A', 'B', 'A', 'C', 'A', 'B', 'D', 'A', 'B')

- Count and print how many times 'A' appears.
- Count and print how many times 'B' appears.
- Take a grade as input from the user and print how many times it appears.

Q8. (index() Method)
Create the tuple: fruits = ('apple', 'banana', 'cherry', 'banana', 'mango', 'apple')

- Find and print the first index of 'banana'.
- Find and print the first index of 'banana' starting the search from index 2.
- Use try-except to safely find the index of 'kiwi' and print "Not found" if it doesn't exist.

Q9. (Immutability of Tuples)
Create a tuple: coordinates = (10, 20)
Demonstrate that tuples are immutable by trying to:

- Change the first element
- Add a new element using append()

Explain what errors occur (in comments). Then show the correct way to modify data by converting the tuple to a list, making changes, and converting back to a tuple.

Q10. (Mini Project - Combined Concepts)
Create a **Student Record Program** using tuples:

- Take input for name, roll number, 3 subject marks.
- Store all information in a tuple (use packing).
- Print the complete record.
- Unpack the tuple and print each value with proper labels.
- Use count() to check how many times a particular mark appears (if any).
- Create a nested tuple to store multiple student records (at least 2 students) and access specific values.

Add meaningful comments and use proper formatting.